Exercise-2

<u>Problem statement</u>: Implementing Ecommerce platform search function

<u>Source code</u>:

```
using System;
class Product
{
    public int ProductId;
    public string ProductName;
    public string Category;
    public Product(int productId, string productName, string category)
    {
        ProductId = productId;
        ProductName = productName;
        Category = category;
    }
    public override string ToString()
    {
        return "Product ID: " + ProductId + ", Name: " + ProductName + ", Category: " + Category;
    }
}
class Program
{
    static Product LinearSearch(Product[] products, int targetId)
    {
```

```
    foreach (Product product in products)
    {
        if (product.ProductId == targetId)
        {
            return product;
        }
    }
    return null;
}
static Product BinarySearch(Product[] products, int targetId)
{
    int left = 0;
    int right = products.Length - 1;

    while (left <= right)
    {
        int mid = left + (right - left) / 2;
        if (products[mid].ProductId == targetId)
        {
            return products[mid];
        }
        else if (products[mid].ProductId < targetId)
        {
            left = mid + 1;
        }
```

```
        else
        {
            right = mid - 1;
        }
    }
    return null;
}
static void Main()
{
    Product[] products =
    {
        new Product(101, "Laptop", "Electronics"),
        new Product(102, "Mobile", "Electronics"),
        new Product(103, "Shoes", "Fashion"),
        new Product(104, "Watch", "Accessories"),
        new Product(105, "Book", "Education")
    };

    int targetId = 104;

    Console.WriteLine("Linear Search:");
    Product result1 = LinearSearch(products, targetId);
    if (result1 != null)
        Console.WriteLine(result1);
    else
```

```
            Console.WriteLine("Product Not Found");
        Console.WriteLine();
        Console.WriteLine("Binary Search:");
        Product result2 = BinarySearch(products, targetId);
        if (result2 != null)
            Console.WriteLine(result2);
        else
            Console.WriteLine("Product Not Found");
        Console.WriteLine();
        Console.WriteLine("Linear Search Time Complexity:");
        Console.WriteLine("Best Case: O(1)");
        Console.WriteLine("Average Case: O(n)");
        Console.WriteLine("Worst Case: O(n)");
        Console.WriteLine();
        Console.WriteLine("Binary Search Time Complexity:");
        Console.WriteLine("Best Case: O(1)");
        Console.WriteLine("Average Case: O(log n)");
        Console.WriteLine("Worst Case: O(log n)");
        Console.WriteLine();
        Console.WriteLine("Binary Search is more suitable for fast product search.");
    }
}
```

Output:

Output

```
Linear Search:
Product ID: 104, Name: Watch, Category: Accessories

Binary Search:
Product ID: 104, Name: Watch, Category: Accessories

Linear Search Time Complexity:
Best Case: O(1)
Average Case: O(n)
Worst Case: O(n)

Binary Search Time Complexity:
Best Case: O(1)
Average Case: O(log n)
Worst Case: O(log n)

Binary Search is more suitable for fast product search.
```